Relativity

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Contents

Prerequisite Knowledge

- Basic algebra with square roots, and comfort with the binomial approximation $(1+x)^n \approx 1+nx$ for small x
- The classical Doppler effect and $c = f\lambda$ (see the Spectroscopy & Atomic Physics handout — Section 6 here builds directly on it)
- Escape velocity and gravitational potential energy (see the Celestial Mechanics handout — Section 11 reuses both, and Section 2.4's centripetal-force reasoning resurfaces once more)
- Mass-energy equivalence, $E = mc^2$, as a bare statement (see the Nuclear Physics handout) — here we'll actually derive where it comes from

1 The Two Postulates

Here's some food for thought the next time you turn on the lights: the speed of light doesn't care how fast you're moving. Chase a beam of light in the fastest spacecraft imaginable, and it still recedes from you at exactly $c \approx 3 \times 10^8$ m/s — not c minus your speed, just c, full stop. This single, almost absurd-sounding fact is the seed of everything in this handout.

Einstein didn't propose this out of nowhere. By the late 1800s, Maxwell's equations (which you met in the Electromagnetic Theory & Quantum Physics handout, governing how light propagates) predicted a fixed speed for light, with no mention of any observer's motion at all — unlike every other wave physicists knew, which always travelled at a fixed speed *relative to some medium* (sound relative to air, water waves relative to the water). Physicists spent decades hunting for light's medium, the "luminiferous aether," including a famous 1887 experiment by Michelson and Morley designed to detect the Earth's motion through it. They found nothing: no matter which direction they measured, or what time of year, light arrived at exactly the same speed. Einstein's radical move in 1905 was to simply take this null result at face value, elevate it to a fundamental law of nature, and see where the consequences led — rather than continuing to hunt for a medium that stubbornly refused to show itself.

Special relativity rests on exactly two postulates:

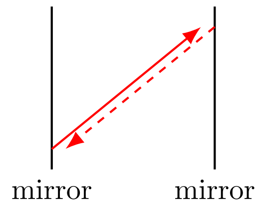
1. **The principle of relativity**: the laws of physics are identical in every inertial (non-accelerating) reference frame. There is no experiment you can perform, sealed inside a windowless lab, that tells you whether you're "at rest" or "moving at constant velocity" — these two situations are physically indistinguishable. (This part isn't new; Galileo already knew it, from a ship's cabin moving smoothly on calm water.)
2. **The constancy of the speed of light**: light travels at the same speed c in every inertial frame, regardless of the motion of the source or the observer.

The first postulate is old news. The second is the genuinely radical one, and the two together are in direct tension with each other in a way that forces almost everything else in this handout. Here's the tension: imagine a spacecraft moving at $0.5c$ relative to Earth, switching on a headlight. Common sense (and ordinary Galilean addition of velocities, the way you'd add a train's speed to a ball thrown from it) says the light should move at $c + 0.5c = 1.5c$ relative to Earth. But the second postulate insists it's still just c. Something familiar has to give — and what gives, it turns out, is our intuitive, unexamined assumptions about time and space themselves.

2 Time Dilation

The Light Clock

Let's build the simplest possible clock: two mirrors, facing each other, a distance L apart, with a single pulse of light bouncing back and forth between them. Each round trip takes $\Delta t_0 = 2L/c$ — call this one "tick." Nothing about a light clock is special except that it lets us track time using something whose speed we know is fixed in every frame: light itself.

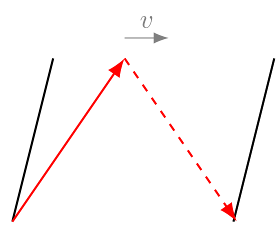


at rest: light bounces straight up and down

Figure 1: A light clock at rest in its own frame. One tick is one round trip, $\Delta t_0 = 2L/c$.

Now put this clock aboard a spacecraft flying past you (the observer on the ground) at speed v. In the spacecraft's own frame, nothing has changed — the light still just bounces straight up and down, ticking at $\Delta t_0 = 2L/c$, exactly as before. This interval, measured in the clock's own rest frame, is called the **proper time**, Δt_0.

But watch what *you*, on the ground, see. From your frame, the whole clock is sliding sideways at speed v while the light bounces, so the light doesn't just go straight up and down — it has to travel diagonally, tracing out a longer, zig-zag path to catch up with the moving mirror each time.



moving: light traces a longer diagonal path

Figure 2: The same clock, seen from the ground, as the spacecraft moves at speed v. Light still travels at c — but now covers more ground per tick.

Deriving the Dilation Factor

Here's the key move: light travels at the *same speed* c in both frames (postulate 2), but it covers a *longer* diagonal path in the ground frame. Since speed is fixed but distance increased, the *time* for one tick, as measured from the ground, Δt, must be longer than Δt_0.

Let's make this precise. In the time $\Delta t/2$ it takes light to go from one mirror to the other (as measured on the ground), the spacecraft itself has moved a horizontal distance $v\Delta t/2$, while the light has travelled a diagonal distance $c\Delta t/2$. These three distances — the vertical separation L, the horizontal drift $v\Delta t/2$, and the diagonal path $c\Delta t/2$ — form a right triangle:

$$\left(\frac{c\Delta t}{2}\right)^2 = L^2 + \left(\frac{v\Delta t}{2}\right)^2.$$

Substituting $L = c\Delta t_0/2$ (from the rest-frame result above) and solving for Δt:

$$\frac{c^2\Delta t^2}{4} - \frac{v^2\Delta t^2}{4} = \frac{c^2\Delta t_0^2}{4} \quad \Longrightarrow \quad \Delta t^2\left(1 - \frac{v^2}{c^2}\right) = \Delta t_0^2,$$

$$\boxed{\Delta t = \gamma\,\Delta t_0}, \qquad \gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}.$$

This factor γ (the **Lorentz factor**) is the single most important number in this entire handout — it will reappear, in one guise or another, in nearly every formula that follows. Notice $\gamma \geq 1$ always, with $\gamma = 1$ exactly when $v = 0$ (no dilation at rest) and $\gamma \to \infty$ as $v \to c$. A moving clock runs *slow*, as judged by a stationary observer — and this isn't some mechanical quirk of light clocks specifically. Since the two postulates apply to *all* physics, not just light, every process aboard the moving spacecraft — heartbeats, chemical reactions, radioactive decay (Nuclear Physics handout, Section 3) — runs slow by the identical factor γ, exactly as seen from the ground.

An Aside: "Proper" Just Means "In Its Own Frame"

The word *proper*, throughout relativity, always means "measured in the frame where the thing being measured is at rest." Proper time is what a clock reads on its own wristwatch; proper length (Section 3) is what a ruler reads when you're holding it still next to the object. Every other observer, in relative motion, measures something *different* — and it's always γ times larger for time, or $1/\gamma$ times smaller for length, never the other way around. Keeping straight *whose* frame a given Δt_0 or L_0 belongs to is the single most common source of sign errors in relativity problems.

Worked Example — Cosmic-Ray Muons

Muons created by cosmic rays hitting the upper atmosphere, roughly 15 km up, have a mean lifetime of $\tau_0 \approx 2.2$ μs in their own rest frame, and travel at $v = 0.998c$. Classically, in τ_0 they'd cover only $v\tau_0 \approx 660$ m before decaying — nowhere near enough to reach the ground. Yet muons are detected at sea level in huge numbers. Time dilation resolves this: from the ground frame, the muon's internal clock runs slow by

$$\gamma = \frac{1}{\sqrt{1 - (0.998)^2}} \approx 15.8,$$

so the muon's decay is delayed to $\gamma\tau_0 \approx 35$ μs *as measured on the ground*, giving it time to cover $v \cdot \gamma\tau_0 \approx 10.5$ km — comfortably enough (especially once you account for the full decay-probability spread rather than just the mean lifetime) to reach detectors at sea level. This isn't a thought experiment; it's one of the most repeated, most direct experimental confirmations of time dilation there is, performed continuously, for free, by the atmosphere above your head.

Stellar Application: Time Dilation Around Compact Objects

Anything orbiting very close to a massive, compact object — a hot spot on the accretion disk of a neutron star or black hole, or a star like S2 swinging through its closest approach to Sagittarius A* — can reach speeds that are a meaningful fraction of c. Time dilation (combined with the gravitational time dilation of Section 10, which typically dominates even closer in) means that a clock riding along with that material genuinely ticks slower than a distant observer's clock — not an illusion of light-travel-time, but an actual difference in the accumulated proper time between the two paths through spacetime. This is precisely the effect that had to be modelled to correctly interpret the orbital period measurements that won the 2020 Nobel Prize in Physics for confirming the supermassive black hole at our galaxy's centre.

3 Length Contraction

Deriving Contraction From Dilation

Now let's ask about *distance* rather than time. Suppose a spacecraft, at speed v, needs to cross a distance L_0 (measured by an observer on the ground, at rest relative to that distance — so L_0 is the *proper length* of this stretch of space). The ground observer simply computes the crossing time as

$$\Delta t = \frac{L_0}{v}.$$

But the spacecraft's own crew, using their own onboard clock, measure their own elapsed proper time for the same journey, $\Delta t_0 = \Delta t/\gamma$ (time dilation, Section 2, applied in reverse — the ground-frame interval is the *dilated* one here, since it's the crew's clock that's moving relative to the ground). From the crew's point of view, they're sitting still while the distance itself rushes past them at speed v, so *they* would naturally compute the length of that same stretch of space as

$$L = v\,\Delta t_0 = v \cdot \frac{\Delta t}{\gamma} = \frac{L_0}{\gamma}.$$

$$\boxed{L = \frac{L_0}{\gamma}.}$$

This is **length contraction**: any length measured by an observer in relative motion along the direction of that length comes out *shorter* than the proper length L_0 measured by an observer at rest relative to it, by exactly the same factor γ that stretched time. Notice the contraction only happens *along the direction of motion* — a spacecraft moving past you horizontally appears squashed along its direction of travel, but its height (perpendicular to the motion) is completely unaffected. This asymmetry is worth sitting with: time dilates for everyone in relative motion regardless of direction, but length only contracts along the one axis that motion actually picks out.

An Aside: Time Dilation and Length Contraction Are the Same Fact

Notice how directly the second derivation leaned on the first: length contraction wasn't an independent postulate, or a separate physical mechanism — it fell straight out of time dilation plus the simple relation $L = v\Delta t$. This is a recurring theme in relativity: there are really only a small handful of independent physical facts (the two postulates), and almost everything else — dilation, contraction, the relativity of simultaneity, the Lorentz transformation itself — is different algebraic *shadows* of those same two facts, viewed from different angles.

Worked Example — The Muon, Revisited

Let's redo the cosmic-ray muon calculation from the muon's own point of view, as a consistency check. In the muon's rest frame, it doesn't experience any time dilation at all (its own clock is, by definition, always "normal" to itself) — so how does it possibly survive the trip down to sea level in only $\tau_0 \approx 2.2\ \mu$s? The answer is length contraction: the 15 km of atmosphere, which is the *ground observer's* proper length, is contracted in the muon's frame to

$$L = \frac{L_0}{\gamma} = \frac{15\ \text{km}}{15.8} \approx 0.95\ \text{km},$$

which the muon, moving at $0.998c$, crosses in just $L/v \approx 3.2\ \mu$s — comfortably within a few decay lifetimes. Both observers agree, in the end, that the muon reaches the ground; they just disagree completely about *why* — one credits a dilated lifetime, the other a contracted atmosphere. This is a genuinely important lesson: different frames can tell wildly different stories about *how* something happened, while still agreeing perfectly on any actual, physically observable outcome.

4 The Lorentz Transformation

Time dilation and length contraction are useful special-case results, but they don't yet tell you how to convert a full *event* — a specific position *and* time, like "the muon decays here, at this moment" — from one frame to another in general. For that we need the full **Lorentz transformation**, the relativistic replacement for the everyday, intuitive Galilean transformation $x' = x - vt,\ t' = t$ (which just says: shift your ruler's origin along with the moving frame, and don't touch time at all — exactly what common sense suggests, and exactly what turns out to be wrong).

Consider two frames, S (the ground) and S' (the spacecraft), with S' moving at speed v along the shared x-axis, and both origins coinciding at $t = t' = 0$. By the principle of relativity, the transformation has to be *linear* (uniform motion should map to uniform motion in either frame) and it has to reduce to the familiar Galilean form when $v \ll c$. A good ansatz, respecting these constraints, is

$$x' = \gamma(x - vt).$$

By the principle of relativity, the reverse transformation (viewing S from S') must have exactly the same form with $v \to -v$:

$$x = \gamma(x' + vt').$$

Now impose the second postulate directly: a light pulse released from the common origin at $t = 0$ must satisfy $x = ct$ in S *and* $x' = ct'$ in S', simultaneously. Substituting $x = ct$ into the first equation and $x' = ct'$ into the second, then eliminating x and x' between them (substitute one into the other) gives, after collecting terms:

$$ct' = \gamma(ct - vt) = \gamma t(c - v), \qquad ct = \gamma(ct' + vt') = \gamma t'(c + v).$$

Multiplying these two equations together:

$$c^2 tt' = \gamma^2 tt'(c - v)(c + v) = \gamma^2 tt'(c^2 - v^2) \quad \Longrightarrow \quad \gamma^2 = \frac{c^2}{c^2 - v^2} = \frac{1}{1 - v^2/c^2},$$

recovering exactly the same γ we found from the light clock — a satisfying consistency check that both routes to relativity (the light-clock argument and this more formal transformation approach) are describing the same underlying structure. Solving the pair of equations above for t' in terms of x and t (a short bit of algebra, substituting back and simplifying) gives the complete Lorentz transformation:

$$\boxed{x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{vx}{c^2}\right).}$$

The Strangest Term in Physics

Stare at that t' equation for a moment, because it's carrying more than it looks like it is. The Galilean version just says $t' = t$ — time is time, everywhere, for everyone, no exceptions. The relativistic version says something genuinely unsettling: t' depends on *where* you are (x), not just on when. Two events that happen *simultaneously* in frame S (same t, different x) do *not*, in general, happen simultaneously in frame S' — this is the **relativity of simultaneity**, arguably the single hardest idea in special relativity to build real intuition for, precisely because it has zero analogue in ordinary, low-speed experience. "At the same time" turns out not to be an absolute, frame-independent statement about the universe at all — it's a statement that only makes sense once you've already picked a specific observer's frame.

Setting $\Delta x = 0$ (a clock that stays at one fixed location in S) recovers time dilation directly: $\Delta t' = \gamma \Delta t$. Setting $\Delta t = 0$ (measuring the two ends of a rod *simultaneously* in S) recovers length contraction: $\Delta x' = \gamma \Delta x$, i.e. $\Delta x = \Delta x'/\gamma$. Both of Sections 2 and 3 are just this one transformation, evaluated along two particular slices.

5 Relativistic Velocity Addition

Go back to the tension we opened with: a spacecraft moving at v relative to Earth fires a probe forward at speed u' (measured in the spacecraft's own frame). What speed u does an observer on Earth measure for the probe? Ordinary Galilean intuition says $u = u' + v$ — just add them. Relativity has to forbid this whenever it would push u past c, so something has to change.

Differentiate the Lorentz transformation (treating dx, dt as small displacements along a trajectory, so $u = dx/dt$ and $u' = dx'/dt'$):

$$dx' = \gamma(dx - v\,dt), \qquad dt' = \gamma\left(dt - \frac{v\,dx}{c^2}\right).$$

Dividing the first by the second, and then dividing both numerator and denominator by dt:

$$u' = \frac{dx'}{dt'} = \frac{dx - v\,dt}{dt - v\,dx/c^2} = \frac{u - v}{1 - uv/c^2}.$$

Solving for u (or equivalently, rerunning the same steps with $v \to -v$ to get the forward-going version) gives the **relativistic velocity addition formula**:

$$\boxed{u = \frac{u' + v}{1 + u'v/c^2}.}$$

Check the two limits that matter most. When both speeds are small compared to c, the denominator's correction term $u'v/c^2$ is negligible, and this collapses right back to the familiar Galilean $u \approx u' + v$ — exactly as it must, since nothing about everyday, slow-speed physics should look broken. But now set $u' = c$ (the probe is itself a beam of light): the formula gives

$$u = \frac{c + v}{1 + v/c} = \frac{c(c + v)}{c + v} = c,$$

regardless of v. No matter how fast the spacecraft is already moving, adding light to its speed still gives exactly c — the formula automatically enforces the second postulate, built directly into its own algebra. You can never combine two sub-light speeds and cross c; the denominator always conspires to hold the result just under the ceiling.

Stellar Application: Apparent Superluminal Motion in AGN Jets

Here's a genuinely striking observational puzzle relativity resolves. Very-long-baseline radio observations of jets from active galactic nuclei (material blasted out from the region around a supermassive black hole) sometimes show blobs of material apparently moving *faster than light* across the sky — a direct measurement violation of everything in this handout, or so it seems at first glance.

The resolution isn't that anything actually exceeds c; it's a geometric illusion caused by the finite travel time of light itself. Suppose a blob moves at speed v (true speed, always $< c$) at a small angle θ to our line of sight, covering a transverse (across-the-sky) distance $D = vt\sin\theta$ in a true time t. But because the blob is also moving *toward* us throughout this motion, it's closing part of the light-travel-time gap as it goes, so the light carrying the "end" of this motion doesn't have to travel as far to reach us as the light carrying the "start." The *observed* elapsed time is compressed to

$$t_{\text{obs}} = t - \frac{vt\cos\theta}{c} = t\left(1 - \frac{v}{c}\cos\theta\right),$$

so the *apparent* transverse velocity, D/t_{obs}, works out to

$$v_{\text{app}} = \frac{v\sin\theta}{1 - (v/c)\cos\theta}.$$

For v close to c and a small, favourably-aligned θ, this can genuinely exceed c — not because anything physical moved faster than light, but purely because the "clock" we're using (light-arrival time) is itself being distorted by the source's own near-light-speed motion toward us. This is precisely the phenomenon observed in jets like that of the quasar 3C 279, and it's simultaneously a beautiful confirmation that jets really do move at close to c, and a cautionary tale about the difference between what you *measure* and what's actually *happening*.

6 The Relativistic Doppler Effect

The classical Doppler effect (Spectroscopy & Atomic Physics handout, Section 5) came purely from light-travel-time bookkeeping: a receding source's successive wave crests each have slightly farther to travel, stretching the observed wavelength. That derivation quietly assumed time ticks the same way for source and observer. It doesn't — so the relativistic version has to fold time dilation in as well.

Consider a source moving directly away from an observer at speed v, emitting light of proper frequency f_0 (measured in the source's own rest frame). Two effects combine:

1. **Classical light-travel-time stretching**, exactly as derived in the Spectroscopy handout: over one period, the source retreats an extra $v\Delta t$, stretching each observed period by a factor $(1 + v/c)$.
2. **Time dilation**: the source's own clock — and hence the rate at which it emits wave crests in the first place — is already running slow, by the usual factor γ, as judged from the observer's frame.

Multiplying these two stretching factors together, the observed period is $T_{\text{obs}} = \gamma(1 + v/c)\,T_0$, so the observed frequency is

$$f_{\text{obs}} = \frac{f_0}{\gamma(1 + v/c)} = f_0\sqrt{1 - \frac{v^2}{c^2}} \cdot \frac{1}{1 + v/c} = f_0\sqrt{\frac{(1 - v/c)(1 + v/c)}{(1 + v/c)^2}},$$

$$\boxed{f_{\text{obs}} = f_0\sqrt{\frac{1 - v/c}{1 + v/c}}} \qquad \text{(source receding).}$$

For a source approaching, simply flip the sign of v. Two features are worth noting explicitly. First, when $v \ll c$, expanding the square root to first order recovers exactly the classical result

$\Delta f/f \approx -v/c$ from the Spectroscopy handout, as it must. Second — and this has no classical counterpart at all — there is a genuine **transverse Doppler effect**: even a source moving *purely across* your line of sight (so the classical light-travel-time argument predicts zero shift at all) is still redshifted, by the pure time-dilation factor $1/\gamma$, because its clock is running slow regardless of which direction it happens to be moving.

Stellar Application: Redshift, Revisited

The Spectroscopy & Atomic Physics handout flagged an important subtlety and deferred it here: that cosmological redshift (the redshift of distant galaxies powering Hubble's law) is technically *not* this kind of Doppler shift at all, but a stretching of the light's wavelength caused by the expansion of space itself while the light is in transit. For nearby galaxies, at low redshift, the two effects are numerically almost indistinguishable, and $z \equiv \Delta\lambda/\lambda \approx v/c$ works as a fine approximation either way. But this relativistic Doppler formula *is* exactly the right tool whenever a genuine, local, high-speed motion is involved — a star's spectral lines as it swings through a tight binary at a meaningful fraction of c (rare, but it happens near compact objects), or light emitted from close to the base of a relativistic jet (Section 5's application, revisited) — while the cosmological case needs general relativity's description of an expanding spacetime, a genuinely different piece of physics wearing a very similar-looking formula.

7 Relativistic Momentum and Energy

Why Ordinary Momentum Breaks

Ordinary momentum, $p = mv$, quietly assumes that mass is a fixed, frame-independent property and that velocities simply add. We've just spent four sections showing that velocities emphatically do *not* simply add. It shouldn't be surprising, then, that $p = mv$ needs fixing too — if it didn't, momentum conservation (one of the most sacred laws in all of physics) would come out different in different frames, violating the very first postulate.

Working through a specific collision in two different frames (a calculation we'll only sketch, since the full algebra is lengthy but adds little physical insight beyond the result), the correct, frame-consistent definition of momentum turns out to be

$$\boxed{p = \gamma m v.}$$

Notice the mass m appearing here is the ordinary, frame-independent **rest mass** — what you'd measure for the object sitting still in front of you. All the velocity-dependence has been absorbed into γ, which now appears explicitly attached to the object's speed, growing without bound as $v \to c$. This is precisely why nothing with mass can ever actually reach c: it would take an infinite momentum (and, as we're about to see, infinite energy) to get there.

Deriving $E = mc^2$

Energy and momentum, in relativity, turn out to be two faces of a single four-dimensional object (a preview of the "spacetime" unification in Section 8), and the cleanest way to see the mass-energy relationship is through the relativistic kinetic energy. Work is force times distance, and force is still the rate of change of momentum, $F = dp/dt$, even relativistically (this part of Newton's second law survives completely intact — it's only $p = mv$ itself that needed revising). The kinetic energy delivered by accelerating a mass from rest up to speed v is

$$KE = \int F\,dx = \int \frac{dp}{dt}\,dx = \int v\,dp.$$

Substituting $p = \gamma mv$ and integrating by parts (a standard but slightly involved calculus exercise, using $d(\gamma v)$ worked out via the chain rule and simplified using $\gamma^2(1 - v^2/c^2) = 1$) yields, after the dust settles,

$$KE = (\gamma - 1)mc^2.$$

This is a genuinely remarkable result on its own: even before assigning any deeper meaning to it, notice that expanding γ for $v \ll c$ via the binomial approximation, $\gamma \approx 1 + \frac{1}{2}\frac{v^2}{c^2}$, gives

$$KE \approx \left(1 + \frac{1}{2}\frac{v^2}{c^2} - 1\right) mc^2 = \frac{1}{2}mv^2,$$

recovering the familiar classical kinetic energy exactly, as any correct relativistic formula must in the low-speed limit. But now look at the *full* expression, $KE = (\gamma - 1)mc^2 = \gamma mc^2 - mc^2$, and notice it naturally splits into two pieces: a term γmc^2 that depends on speed, and a fixed term mc^2 that doesn't go away even when $v = 0$ (where $\gamma = 1$ and $KE = 0$ exactly, as expected). Einstein's interpretive leap was to take this leftover, ever-present term seriously as a form of energy in its own right — the energy an object has purely by virtue of *having mass at all*, even sitting perfectly still. Defining total energy as

$$\boxed{E = \gamma mc^2 = KE + mc^2},$$

the object's **rest energy** is $E_0 = mc^2$ — exactly the equation you've already met in the Nuclear Physics handout, now derived rather than simply asserted, and now visibly just the $v = 0$ special case of a broader statement about energy and motion together.

The Energy-Momentum Relation

One more relation is worth having, since it's often more directly useful than either E or p alone — especially for massless particles like photons, where $E = \gamma mc^2$ is a $0/0$ indeterminate mess (since $m = 0$ and $\gamma \to \infty$ for a particle already moving at c). Starting from $E = \gamma mc^2$ and $p = \gamma mv$, eliminate γ between them (square both, form $E^2 - (pc)^2$, and use $\gamma^2(1 - v^2/c^2) = 1$ once more to simplify):

$$\boxed{E^2 = (pc)^2 + (mc^2)^2.}$$

This single equation cleanly covers every case at once. For a particle at rest ($p = 0$), it collapses to $E = mc^2$, the rest energy. For a massless particle like a photon ($m = 0$), it gives the clean relation $E = pc$ — light carries momentum, and quite a lot of it for its energy, completely without needing any mass to do so. And for anything in between, it correctly interpolates, reducing to the familiar $KE \approx p^2/2m$ in the low-speed, non-relativistic limit (worth checking as an exercise: expand $E = mc^2\sqrt{1 + (p/mc)^2}$ via the binomial approximation for small p/mc and subtract off the rest energy).

Stellar Application: Mass Defect, Now With a Real Derivation

The Nuclear Physics handout introduced binding energy via $BE = \Delta m\, c^2$, treating $E = mc^2$ as a given tool. We can now see exactly why that bookkeeping works: a bound nucleus, at rest, has a total energy Mc^2 where M is its measured rest mass — but this same energy could equally be computed by adding up the rest energies of its separate, individual nucleons, $\sum m_i c^2$, minus whatever energy had to be *released* to bind them together in the first place. Since energy is conserved, the released binding energy has to show up as *missing* rest energy in the final, bound object — exactly the mass defect $\Delta m = \sum m_i - M$. There's nothing mysterious about mass "turning into" energy; it's the single conservation law $E = \gamma mc^2$, applied consistently before and after the nucleus assembles itself.

An Aside: Why Photons Have Momentum Despite Having No Mass

It's worth pausing on how strange $E = pc$ (for a massless particle) really is, compared to classical intuition, where momentum was always "mass times velocity" and a massless object seemed like it ought to carry nothing at all. Relativity says otherwise: momentum is fundamentally about how a particle's energy is distributed *directionally* through spacetime, not specifically about mass in motion. This is exactly what makes solar sails and radiation pressure possible — sunlight, despite consisting of massless photons, delivers a real, measurable push, precisely because $p = E/c$ is nonzero for every photon that strikes a surface and is absorbed or reflected.

8 The Spacetime Interval

Every quantity we've derived so far — Δt, L, individual velocities — changes when you switch frames. It's worth asking whether *anything* stays the same for every observer, the way ordinary distance Δx did in pre-relativistic physics. There is exactly one such quantity, and finding it is what finally lets us stop thinking of time and space as two separate things.

Define the **spacetime interval** between two events as

$$\Delta s^2 \equiv c^2 \Delta t^2 - \Delta x^2.$$

Substituting the Lorentz transformation for $\Delta x'$ and $c\Delta t'$ and expanding, every cross-term cancels and every remaining term collapses back to exactly $c^2\Delta t^2 - \Delta x^2$ (a short algebraic exercise worth doing once by hand) — meaning Δs^2 comes out *identical* in every inertial frame, even though Δt and Δx individually do not. This is the deep structural fact hiding underneath everything derived so far: space and time separately are not absolute, but a specific combination of the two — the spacetime interval — is. This is precisely the sense in which relativity "merges" space and time into a single four-dimensional **spacetime**: it's the arena in which Δs^2, not Δx or Δt on their own, is the physically meaningful, observer-independent notion of "separation."

The sign of Δs^2 sorts every possible pair of events into one of three physically distinct categories:

- $\Delta s^2 > 0$ (**timelike**): a slower-than-light signal, or an ordinary massive object, could travel between the two events. Every observer agrees on their time-ordering — cause can precede effect, unambiguously, in every frame.
- $\Delta s^2 = 0$ (**lightlike**, or null): only light itself can connect the two events.
- $\Delta s^2 < 0$ (**spacelike**): no signal, of any kind, can connect the two events — they're too far apart in space, given how little time separates them. Different observers can even disagree on which of the two events happened first, without any contradiction, precisely *because* no cause-and-effect relationship between them was ever possible in the first place.

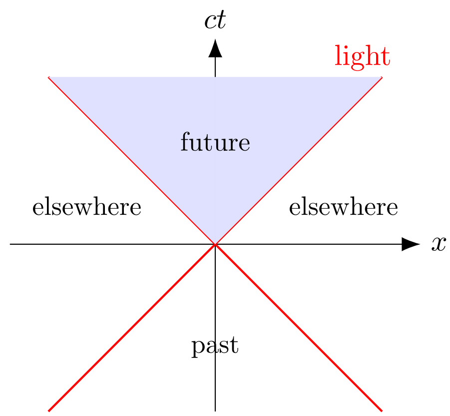


Figure 3: A spacetime (light-cone) diagram. Events inside the shaded cone are timelike-separated from the origin (reachable by a sub-light signal); events outside are spacelike-separated (unreachable, and hence with no absolute time-order relative to the origin).

9 General Relativity: The Equivalence Principle

Everything so far — special relativity — describes physics between observers moving at *constant* relative velocity. It says nothing yet about gravity, or about acceleration. Getting there took Einstein another decade, and it started from a thought experiment he later called "the happiest thought of my life."

Picture yourself sealed in a windowless elevator, far out in deep space, with the elevator's engine firing to accelerate you "upward" at exactly $g = 9.8$ m/s^2. Drop a ball, and it falls to the floor, accelerating at g, exactly as it would in an elevator sitting still on Earth's surface. Einstein's insight was that there is no experiment — none whatsoever, performed entirely inside the sealed elevator — that can distinguish between these two situations: uniform acceleration through empty space, and sitting still in a uniform gravitational field. This is the **equivalence principle**, and it elevates something that Newtonian mechanics had always treated as a slightly suspicious coincidence — that gravitational mass (the m in $F = GMm/r^2$, Celestial Mechanics handout, Section 1) and inertial mass (the m in $F = ma$) always come out numerically identical, for every object ever tested, to arbitrary precision — into the foundational postulate of an entire theory of gravity.

If acceleration and gravity are truly indistinguishable, then anything that happens under acceleration (which we can analyse using special relativity, since an accelerating frame can be approximated, instant by instant, as a sequence of momentarily-matched inertial frames) must *also* happen under gravity. This single idea, followed rigorously, is the entire engine driving general relativity's key predictions — including the next two sections, both derived from exactly this reasoning rather than from any new postulate.

10 Gravitational Time Dilation and Redshift

A Photon Climbing Out of the Accelerating Elevator

Go back to the accelerating elevator, of height h, accelerating upward at g. A photon of frequency f_0 is emitted from the floor, travelling straight up toward the ceiling. In the brief time $t \approx h/c$ it takes light to cross the elevator, the *ceiling* (where the photon will be received) has picked up an extra upward speed of $\Delta v = gt \approx gh/c$, compared to the floor at the moment of emission. So the receiver, at the ceiling, is now moving *away* from the source at this extra speed Δv, and by the ordinary (low-speed, so we can use the classical formula) Doppler effect, detects a redshifted frequency:

$$\frac{\Delta f}{f_0} \approx -\frac{\Delta v}{c} = -\frac{gh}{c^2}.$$

By the equivalence principle, this exact same shift must happen for light climbing *upward*, against gravity, through a height h in a genuine gravitational field of local strength g — even with the elevator's engine switched off entirely, sitting at rest on a planet's surface. Light climbing out of a gravitational well loses energy (and hence frequency) to the climb, exactly the way a thrown ball loses kinetic energy climbing against gravity — this is **gravitational redshift**.

Generalizing from a uniform, local g to a real, varying gravitational field, the total shift between two points at gravitational potentials Φ_1 and Φ_2 (recall $\Phi = -GM/r$ for a point mass, so Φ becomes more negative closer in) works out to

$$\boxed{\frac{\Delta f}{f} \approx \frac{\Phi_1 - \Phi_2}{c^2},}$$

or equivalently, for light climbing away from a spherical mass M from radius r out to a very distant observer ($\Phi \to 0$):

$$\boxed{\frac{f_{\text{observed}}}{f_{\text{emitted}}} \approx 1 - \frac{GM}{rc^2}.}$$

Light emitted deeper in a gravity well (smaller r, more negative Φ) arrives redshifted; light falling *into* a gravity well is correspondingly blueshifted by exactly the reverse factor. Note this is a genuinely separate effect from both the ordinary Doppler shift (Section 6 and the Spectroscopy handout) and cosmological redshift — all three can be present simultaneously in a real astrophysical spectrum, and disentangling them is often the actual point of an IOAA-style problem.

Worked Example — Redshift From a White Dwarf's Surface (IOAA style)

A white dwarf has mass $M = 1.0\,M_\odot \approx 2 \times 10^{30}$ kg and radius $R \approx 6{,}000$ km ($\approx 0.86 R_\oplus$, typical for a white dwarf). Light of rest-frame wavelength $\lambda_0 = 500$ nm, emitted from its surface, is observed from far away. The fractional redshift is

$$\frac{\Delta\lambda}{\lambda} \approx \frac{GM}{Rc^2} = \frac{(6.67 \times 10^{-11})(2 \times 10^{30})}{(6 \times 10^6)(3 \times 10^8)^2} \approx 2.5 \times 10^{-4},$$

giving a wavelength shift of $\Delta\lambda \approx 0.12$ nm — small, but genuinely measurable (and historically, gravitational redshift measurements of Sirius B, a white dwarf, were among the earliest observational tests of general relativity). Notice this is a purely gravitational effect, present even for a perfectly stationary white dwarf, with no orbital or recessional Doppler shift involved at all.

Stellar Application: GPS Satellites, and Neutron Star Surface Gravity

Gravitational time dilation isn't just an exotic astrophysical curiosity — it's load-bearing engineering. GPS satellites orbit at an altitude where Earth's gravity is weaker than at the surface, so their clocks run *fast* relative to ground clocks (a gravitational blueshift, in this case) by about 45 microseconds per day, partly offset by an opposing *special* relativistic time-dilation effect (Section

2) from their orbital speed, which runs their clocks *slow* by about 7 microseconds per day. The net effect, about 38 microseconds per day, sounds tiny, but left uncorrected it would accumulate into several kilometres of positioning error within a single day — so GPS satellite clocks are deliberately built to run at a slightly different rate, correcting for exactly this combination of both effects derived in this handout, before launch.

At the far, extreme end of the same physics: for a neutron star ($M \approx 1.4M_\odot$, $R \approx 10$ km), GM/Rc^2 climbs to roughly 0.15–0.2 — no longer a small correction at all, but a substantial fraction of the object's entire rest energy, and precisely the kind of regime where measuring a spectral line's gravitational redshift becomes a genuine tool for pinning down a neutron star's mass-to-radius ratio directly, entirely independent of any orbital dynamics.

11 The Schwarzschild Radius and Black Holes

A Newtonian Coincidence That Turns Out to Be Exactly Right

Here's a calculation you're already fully equipped to do, using nothing but the escape velocity formula from the Celestial Mechanics handout (Section 2.9): $v_{\text{esc}} = \sqrt{2GM/R}$. Ask a slightly reckless question — purely classically, with no relativity at all — at what radius R does the escape velocity from a mass M reach c itself? Setting $v_{\text{esc}} = c$ and solving:

$$R_s = \frac{2GM}{c^2}.$$

This exact calculation was actually carried out, remarkably, by John Michell in 1783 and (independently) Pierre-Simon Laplace shortly after — over a century before relativity existed at all — speculating about "dark stars" so dense that not even light could escape them. Their reasoning was built on a Newtonian foundation that doesn't actually apply to light (light doesn't have a rest mass to plug into $v_{\text{esc}} = \sqrt{2GM/R}$ in the first place, so the derivation is, in a real sense, not rigorously justified). And yet, when Karl Schwarzschild solved Einstein's field equations properly in 1916 for the spacetime around a spherical, non-rotating mass, the radius at which the mathematics genuinely breaks down — the boundary beyond which absolutely nothing, light included, can ever climb back out — comes out to be *exactly* this same formula:

$$\boxed{R_s = \frac{2GM}{c^2}.}$$

This is a striking, celebrated coincidence, not a case of the naive derivation secretly being correct all along — the real general-relativistic reasoning behind R_s is entirely different from Michell and Laplace's, and involves the actual curvature of spacetime itself, well beyond what this handout can build from scratch. But the fact that two completely different chains of reasoning, a century apart, land on the identical formula is exactly the kind of thing worth knowing walked into an exam.

Event Horizons and Black Holes

R_s marks the **event horizon**: the boundary of the region from within which no signal of any kind can ever reach an outside observer, at any point in the entire future. Notice from the gravitational redshift formula (Section 10) what happens as $r \to R_s$: the observed frequency $f_{\text{observed}} = f_{\text{emitted}}(1 - GM/rc^2) = f_{\text{emitted}}(1 - R_s/2r)$ drops toward zero — light emitted from right at the horizon is redshifted into complete invisibility, taking an infinitely long time (as seen from far away) to actually cross it, even though, from the infalling object's own perspective, crossing the horizon is a perfectly unremarkable, finite-time event. This is another genuine case of two frames disagreeing sharply about *when* something happens, exactly the theme Section 4 first introduced with simultaneity — pushed here to its most extreme, observable limit.

An object of mass M becomes a black hole precisely when its actual physical radius shrinks below $R_s(M)$ — and since every known mass has *some* finite R_s, in principle any object, compressed enough, becomes a black hole. In practice, only sufficiently massive stellar cores (Nuclear Physics handout, Section 2's iron-peak endpoint, taken past the neutron-star stage entirely) ever actually get compressed that far.

Worked Example — Schwarzschild Radii, Two Extremes (IOAA style)

For the Sun, $M_\odot \approx 2 \times 10^{30}$ kg:

$$R_s = \frac{2(6.67 \times 10^{-11})(2 \times 10^{30})}{(3 \times 10^8)^2} \approx 3.0 \text{ km}$$

— meaning that if the entire Sun (radius $\approx 7 \times 10^5$ km in reality) were somehow compressed down to a 3 km sphere without losing any mass, it would become a black hole. For Sagittarius A*, the Milky Way's central supermassive black hole, $M \approx 4.3 \times 10^6 M_\odot$:

$$R_s \approx 4.3 \times 10^6 \times 3.0 \text{ km} \approx 1.3 \times 10^7 \text{ km} \approx 0.086 \text{ AU},$$

a scale genuinely comparable to the orbit of Mercury — despite Sagittarius A*'s event horizon being, by astronomical standards, an almost unbelievably compact object for that much mass to be crammed into.

A Brief Word Beyond the Schwarzschild Radius

Two further predictions of general relativity are worth having on hand for IOAA-level problems, even though deriving them properly requires machinery (the full curved-spacetime field equations) well beyond this handout's scope:

- **Light bending**: a light ray passing a mass M at closest distance (impact parameter) b is deflected by an angle $\delta \approx \dfrac{4GM}{bc^2}$ — exactly twice the (wrong) answer a naive Newtonian calculation of a "massless particle" grazing past M would give, with the extra factor of two coming from spacetime curvature itself, not just gravitational attraction. This was the effect measured during the 1919 solar eclipse expedition that made Einstein famous practically overnight, and is the same physics behind gravitational lensing of background galaxies by foreground galaxy clusters.
- **Perihelion precession**: an elliptical orbit (Celestial Mechanics handout, Section 2.5) doesn't quite close on itself in general relativity — the perihelion itself slowly precesses, by an angle per orbit of $\delta\phi \approx \dfrac{6\pi GM}{a(1-e^2)c^2}$, where a and e are the orbit's semi-major axis and eccentricity. For Mercury, this amounts to a famously tiny 43 arcseconds per century, but it was already a known, unexplained discrepancy in the pre-relativity orbital tables of the 19th century — meaning general relativity didn't just predict a new effect, it retroactively explained one that had been sitting, unresolved, in the data for decades.

Stellar Application: Closing the Loop on Core Collapse

The Nuclear Physics handout left off with a massive star's iron core finally exhausting every available source of fusion energy (Section 2's peak of the binding-energy curve) and collapsing. Whether that collapse halts at a neutron star (held up by neutron degeneracy pressure, a genuinely quantum-mechanical effect beyond what either handout derives) or continues all the way down to a black hole depends essentially on whether the collapsing core's mass exceeds a critical threshold (the Tolman-Oppenheimer-Volkoff limit, roughly 2–3 $M_\odot$, analogous in spirit to, but distinct from, the White Dwarfs handout section's Chandrasekhar limit for electron degeneracy). Above that threshold, no known form of pressure — degenerate or otherwise — can win against gravity at all, and the collapse runs away completely, all the way down past R_s. This is, quite literally, the endpoint of the entire chain of physics traced across every handout in this set: fusion up to iron, core collapse, a neutrino burst racing ahead of the light (Nuclear Physics, Section 4), and finally, for the most massive cores, an object so compact that not even that same light, once it tries to leave, is ever seen again.

12 Summary Sheet

Topic	**Result**
Lorentz factor	$\gamma = 1/\sqrt{1 - v^2/c^2}$
Time dilation	$\Delta t = \gamma \Delta t_0$
Length contraction	$L = L_0/\gamma$
Lorentz transformation	$x' = \gamma(x - vt)$, $t' = \gamma(t - vx/c^2)$
Velocity addition	$u = (u' + v)/(1 + u'v/c^2)$
Apparent superluminal motion	$v_{\text{app}} = v \sin\theta/(1 - (v/c)\cos\theta)$
Relativistic Doppler (receding)	$f_{\text{obs}} = f_0\sqrt{(1 - v/c)/(1 + v/c)}$
Relativistic momentum	$p = \gamma m v$
Total energy / rest energy	$E = \gamma mc^2$; $E_0 = mc^2$
Energy-momentum relation	$E^2 = (pc)^2 + (mc^2)^2$; photons: $E = pc$
Spacetime interval (invariant)	$\Delta s^2 = c^2\Delta t^2 - \Delta x^2$
Equivalence principle	gravity $\equiv$ acceleration, locally indistinguishable
Gravitational redshift	$f_{\text{obs}}/f_{\text{emit}} \approx 1 - GM/rc^2$
Schwarzschild radius	$R_s = 2GM/c^2$
Light bending (impact parameter b)	$\delta \approx 4GM/bc^2$
Perihelion precession (per orbit)	$\delta\phi \approx 6\pi GM/[a(1 - e^2)c^2]$